



DTM-CC016A Single Antenna Multi-Channel Transceiver Gateway Module

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上海磐启微电子有限公司

单天线多通道收发网关模组

Overview

DTM-CC016A gateway module is a single-antenna multi-channel transceiver gateway module developed by Shanghai Panqi Microelectronics Co., Ltd. based on the PAN3029 chip. The module has a built-in M4 processor, integrated PA and LNA, and can efficiently process Chirp-IoT data transmission and reception. The module operates in the 470-510MHz frequency band and supports rate adaptation and communication data AES encryption.

Users can drive the module through the SPI interface. The gateway module supports RF data reception, data transmission, parameter configuration and other interfaces, which is convenient for users to use. The gateway module can be widely used in one-to-many star networks, supports large-capacity terminal node networks, supports ChirpLAN Mode A/B/C protocols, supports multiple application scenarios, etc.

Key Features

- Operating frequency band: 470~510MHz
- Modulation: Chirp-IoT
- Channels: 8RX+1TX, half-duplex
- Communication rate: Support 8 Chirp-IoT data channels, adaptive
- Receiving sensitivity: -142dBm@SF12 BW62.5kHz
- Maximum transmit power: 28dBm
- Encryption mode: AES hardware encryption
- Communication interface: SPI
- RSSI detection and output
- ChirpLAN Mode A/B/C protocols

Application Areas

- Smart city
- Smart rubber tapping
- Smart farm
- Smart power and other industrial scenarios



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1 Hardware

1.1 Physical Picture



Figure 1-1 Module physical picture



Figure 1-2 Gateway physical picture (Raspberry Pi + baseboard + module)

The gateway module has 8 built-in PAN3029, including 1 pre-stage and 7 post-stage. The pre-stage can be used as both receiving and sending, while the post-stage can only be used as receiving.

1.3 Module dimensions

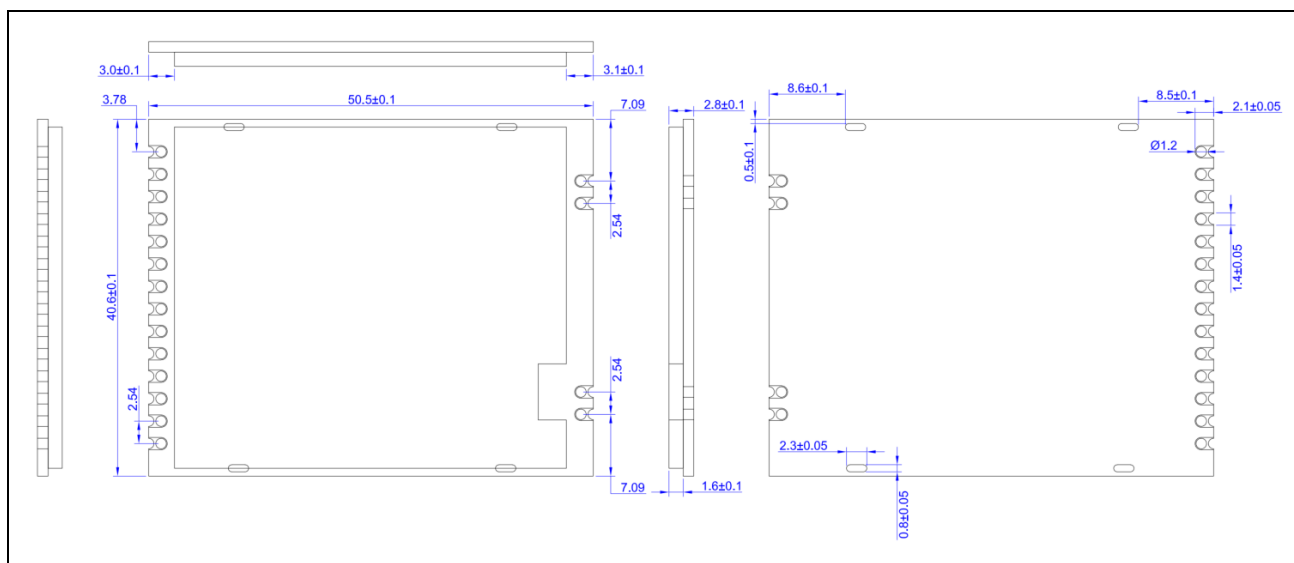


Figure 1-4 Gateway module dimensions

1.4 Module Pin Definition

Port	Description
SPI communication port	1 SPI CSN1_0, MOSI1, MISO1, maximum rate 10MHz
IO1	Custom function pin
Reset	Reset port, RST, 0 active
Programming port	SWD, SWCLK
Serial communication port	UART TX, RX
Power supply	VCC 1PIN, GND 2PIN
Power control	PWR_EN module power switch control pin; 0 - off, 1 - on

1.5 Module baseboard interface definition

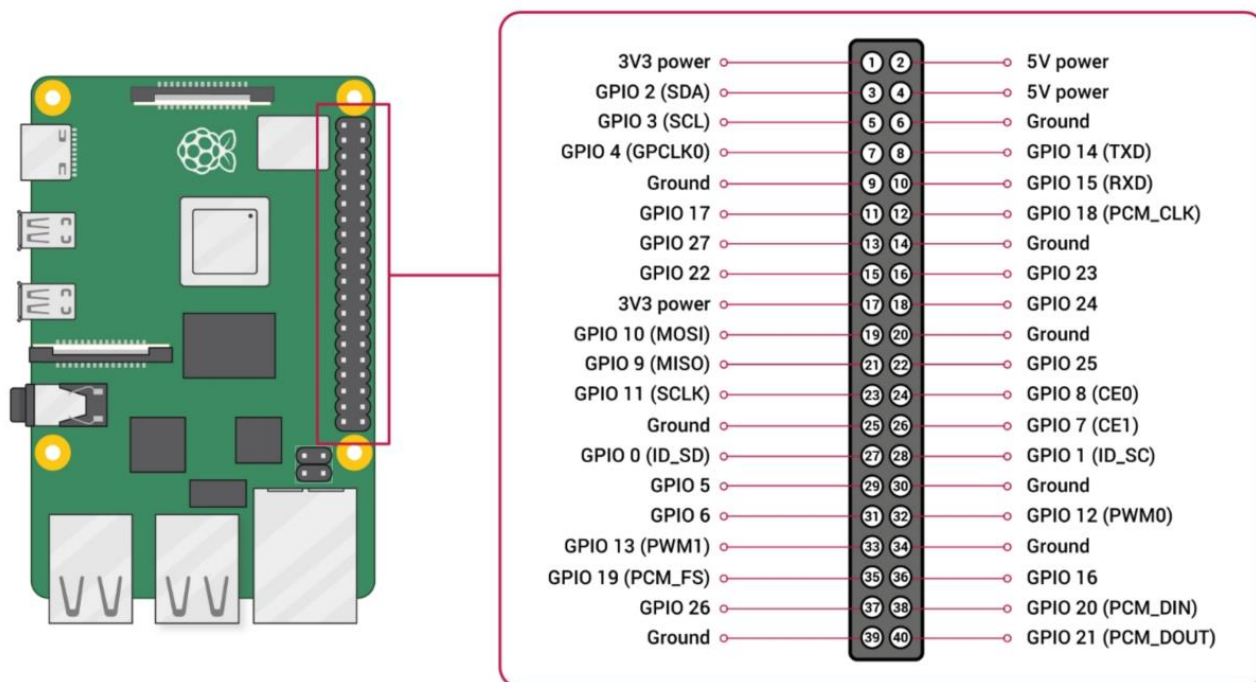


Figure 1-5 Gateway module baseboard pin diagram

1.6 Electrical parameters

Parameter	Description
Power supply voltage	Nominal voltage 5V/2A, input range (4.5V-18V)
Receive current	72mA
Transmit current	870mA@28dBm,470MHz
RF interface	50Ω

1.7 Physical properties

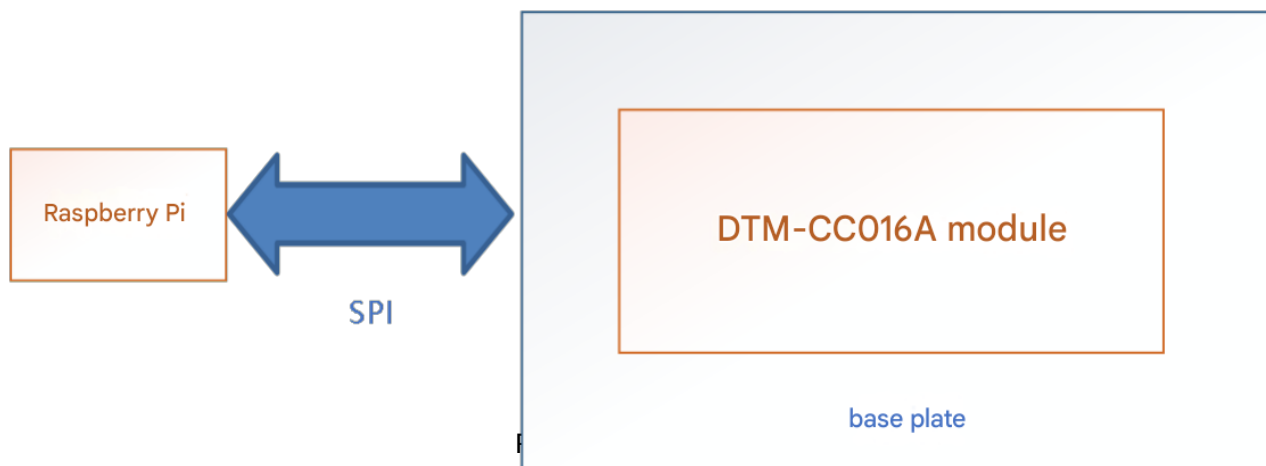
Parameter	Description
Size	50.5*40.6mm
Package	Stamp hole
Temperature	-40~85°C (operating), -55~105°C (storage)
Humidity	5~95% (no condensation)

1.8 RF Parameters

Parameter	Description
Output power	Maximum 28dBm
Bandwidth	62.5kHz, 125kHz, 250kHz, 500kHz
Spreading factor	SF5~SF12
Receiver sensitivity	-142dBm@SF12 BW62.5kHz
Frequency range	470~510MHz
Channels	8 ChirpLoT receive channels, 8RX+1TX half-duplex

2 Module Protocol

The gateway module provides an SPI interface to the outside world. The module acts as an SPI slave. Our company provides Raspberry Pi master control operation examples and gateway module programs. The open source address is: <https://gitee.com/chirplan>



The SPI communication protocol between the module and the main control MCU (Raspberry Pi) is as follows:

(1) The Raspberry Pi obtains the RF data received by the module. The received data content format is as follows:

接收数据区	数据段起始地址	数据存储 (buffer size)	说明
区域说明	0	buf[0]接收区数据大小低8位	
接收数据FIFO检查区	1	buf[1]接收区数据大小高8位	SPI主机读取. 从机判断等于0时将数据放置接收区, 然后将数据大小写至该区域
接收数据FIFO数据区	2	0xxx	Packet数量(协议示例此值为2)
	3	payload_size	payload长度
	0+4	PAYLOAD	PAYLOAD DATA
	...		
	...		
	...		
	payload_size+3	Channel	信道号 (0-7)
	payload_size+4	SF[7:4]CR[3:1]CRC_EN[0]	扩频因子5-12, CR:1-4, CRC:0-1
	5+payload_size	BW[7:4]LDR[0]	带宽 (6\7\8\9分别代表62.5, 125, 250, 500) LDR0/1
	6+payload_size	RSSI	信号质量
	7+payload_size	payload_size	
		PAYLOAD	
		Channel	
		SF[7:4]CR[3:1]CRC_EN[0]	
		BW[7:4]LDR[0]	
		RSSI	
		CRC_L	CRC16低8位(上面所有字节)
		CRC_H	CRC16高8位

Buffer size

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(2) The master control needs to send data through the gateway module.
The protocol format is as follows:

Receive data area				
Area description	Data segment starting address	Data storage (buffer size)	illustrate	
Receive data FIFO check area	0	buf[0] receiving area data size lower 8 bits	The DT master reads, and when the slave judges that it is equal to 0, it places the data in the receiving area, and then writes the data in both upper and lower cases to this area.	
	1	buf[1] receiving area data size high 8 bits		
Receive data FIFO data area	2	0xxx	Packet number (in the protocol example, this value is 2)	
	3	payload_size	payload length	
	0+4	PAYLOAD	PAYLOAD DATA	
	...			
	...			
	...			
	payload_size+3			
	payload_size+4	Channel	Channel number (0-7)	
	5+payload_size	SF[7:4]CR[3:1]CRC_EN[0]	Spreading factor 5-12, CR:1-4, CRC:0-1	
	6+payload_size	BW[7:4]LDR[0]	Bandwidth (617 819 represent 62.5, 125, 250, 500 respectively) LDR0/1	
	7+payload_size	RSSI	signal quality	
		payload_size	Buffer size	
		PAYLOAD		
		Channel		
		SF[7:4]CR[3:1]CRC_EN[0]		
		BW[7:4]LDR[0]		
		RSSI		
		CRC_L	CRC16 lower 8 bits (all bytes above)	
		CRC_H	CRC16 high 8 bits	

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(3) The master controller configures parameters for the gateway module using the following protocol format:

Parameter configuration instruction	data storage	description
0	CMD_CONFIG(0x0A)	Parameter Configuration area
1	buffer_size[0]	Length low byte of the entire packet
2	buffer_size[1]	The length of the entire packet high byte
3	p_center_freq	Pre-stage intermediate frequency (reference value 860K) (default value 1000000)
4		
5		
6		
7	filter_bw	Filter bandwidth (0:500K, 1:1M, 2:2M) (default 2)
8	AGC_ENABLE	Is AGC enabled? 1: Enable, 0: Disable (default value 0)
9	AGC_REG1	Fixed gain 1 (default value 0xd6)
10	AGC_REG2	Fixed gain 2 (default value 0xf9)
11	AGC_REG3	Fixed gain 3 (0x3c)
12	SF0[7:4]CR0[3:1]CRC_EN0[0]	Channel 0 selective parameters SF: 5-12, Smart Search (0:SF5-8, 1:6-9, 2:7-10, 3:8-11, 4:9-12) CR: 1-4, CRC: 0-1 (Default value 0x79)
13	BW0[7:4]LDR0[0]	(Default value 0x70)
14	FREQ0	Channel 0 frequency, unit: Hz, low byte first (default value 484500000)
15		
16		
17		
18	SF1[7:4]CR1[3:1]CRC_EN1[0]	(Default value 0x79)
19	BW1[7:4]LDR1[0]	(Default value 0x70)
20	FREQ1	Channel 1 frequency, unit: Hz, low byte first (default value 484200000)
21		
22		
23		
24	SF2[7:4]CR2[3:1]CRC_EN2[0]	(Default value 0x79)
25	BW2[7:4]LDR2[0]	(Default value 0x70)
26	FREQ2	Channel 2 frequency, unit: Hz, low byte first (default value 483900000)
27		
28		
29		
30	SF3[7:4]CR3[3:1]CRC_EN3[0]	(Default value 0x79)
31	BW3[7:4]LDR3[0]	(Default value 0x70)
32	FREQ3	Channel 3 frequency, unit: Hz, low byte first (default value 483600000)
33		
34		
35		
36	SF4[7:4]CR4[3:1]CRC_EN4[0]	(Default value 0x79)
37	BW4[7:4]LDR4[0]	(Default value 0x70)
38	FREQ4	Channel 4 frequency, unit: Hz, low byte first (default value 483300000)
39		
40		
41		
42	SF5[7:4]CR5[3:1]CRC_EN5[0]	(Default value 0x79)
43	BW5[7:4]LDR5[0]	(Default value 0x70)
44	FREQ5	Channel 5 frequency, unit: Hz, low byte first (default value 483000000)
45		
46		
47		
48	SF6[7:4]CR6[3:1]CRC_EN6[0]	(Default value 0x79)
49	BW6[7:4]LDR6[0]	(Default value 0x70)
50	FREQ6	Channel 6 frequency, unit: Hz, low byte first (default value 482700000)
51		
52		
53		
54	SF7[7:4]CR7[3:1]CRC_EN7[0]	(Default value 0x79)
55	BW7[7:4]LDR7[0]	(Default value 0x70)
56	FREQ7	Channel 7 frequency, unit: Hz, low byte first (default value 482400000)
57		
58		
59		
60	CRC_L	CRC16 lower 8 bits (all bytes above)
61	CRC_H	CRC16 high 8 bits

Buffer size

Revision History

Version	Date	Content
1.0	2023.09	First edition

Documentation

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